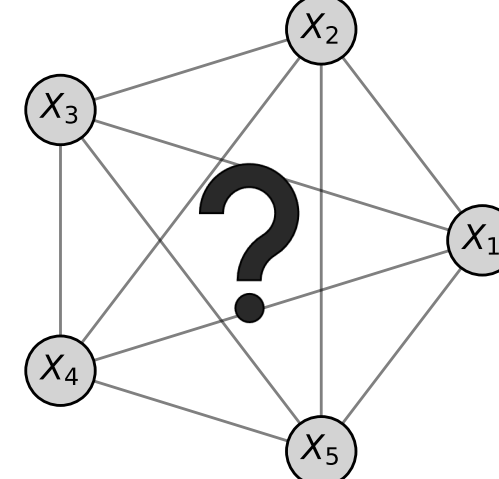




Problem statement

- Learn causal relations between variables as a directed, acyclic graph (DAG) from observational and interventional data
- Assumptions: interventions are sparse (only one variable at a time), soft (distribution over values), perfect (new distribution independent of original parents), and available for all variables.
- Continuous-optimization methods are promising due to their efficiency, but acyclicity needs to be ensured by constrained optimization methods which are slow and sensitive to hyperparameters, or regularizers without guarantess

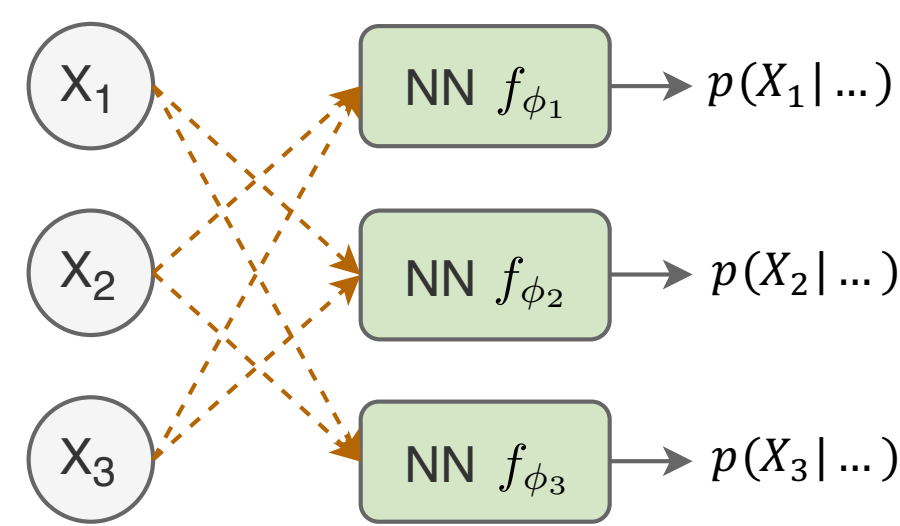


ENCO – Efficient Neural Causal Discovery

- Optimize likelihoods of edges based on how well graphs generalize from observations to interventions
- Split parameters into two groups: edge existence γ and orientation θ for better control on gradients
- Probability of edges: $p(X_i \rightarrow X_j) = \sigma(\gamma_{ij}) \cdot \sigma(\theta_{ij})$
- Fit observational distributions by neural networks, and evaluate different graphs on likelihood-based score function without acyclicity constraint:

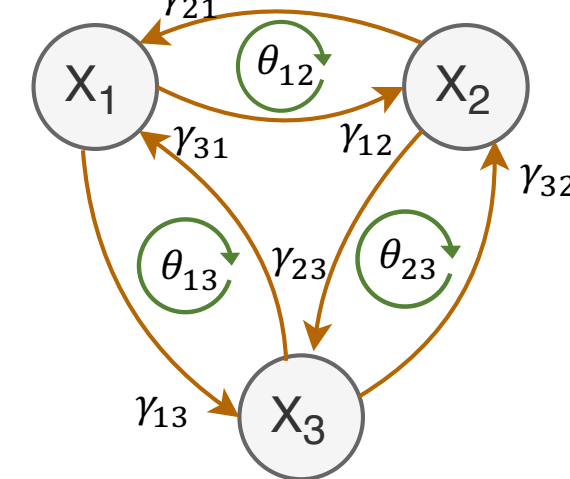
$$\tilde{\mathcal{L}} = \mathbb{E}_{\hat{I} \sim p_I(I)} \mathbb{E}_{\hat{p}_I(\mathbf{X})} \mathbb{E}_{p_{\gamma, \theta}(C)} \left[\sum_{i=1}^N \mathcal{L}_C(X_i) \right] + \lambda_{\text{sparse}} \sum_{i=1}^N \sum_{j=1}^N \sigma(\gamma_{ij}) \cdot \sigma(\theta_{ij})$$

Distribution fitting

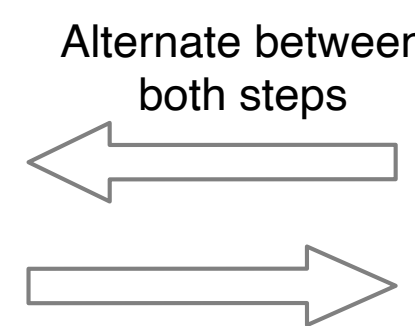


→ Fit NNs to conditional, observational distributions

Graph fitting



→ Learn edge and orientation parameters based on fitted distributions

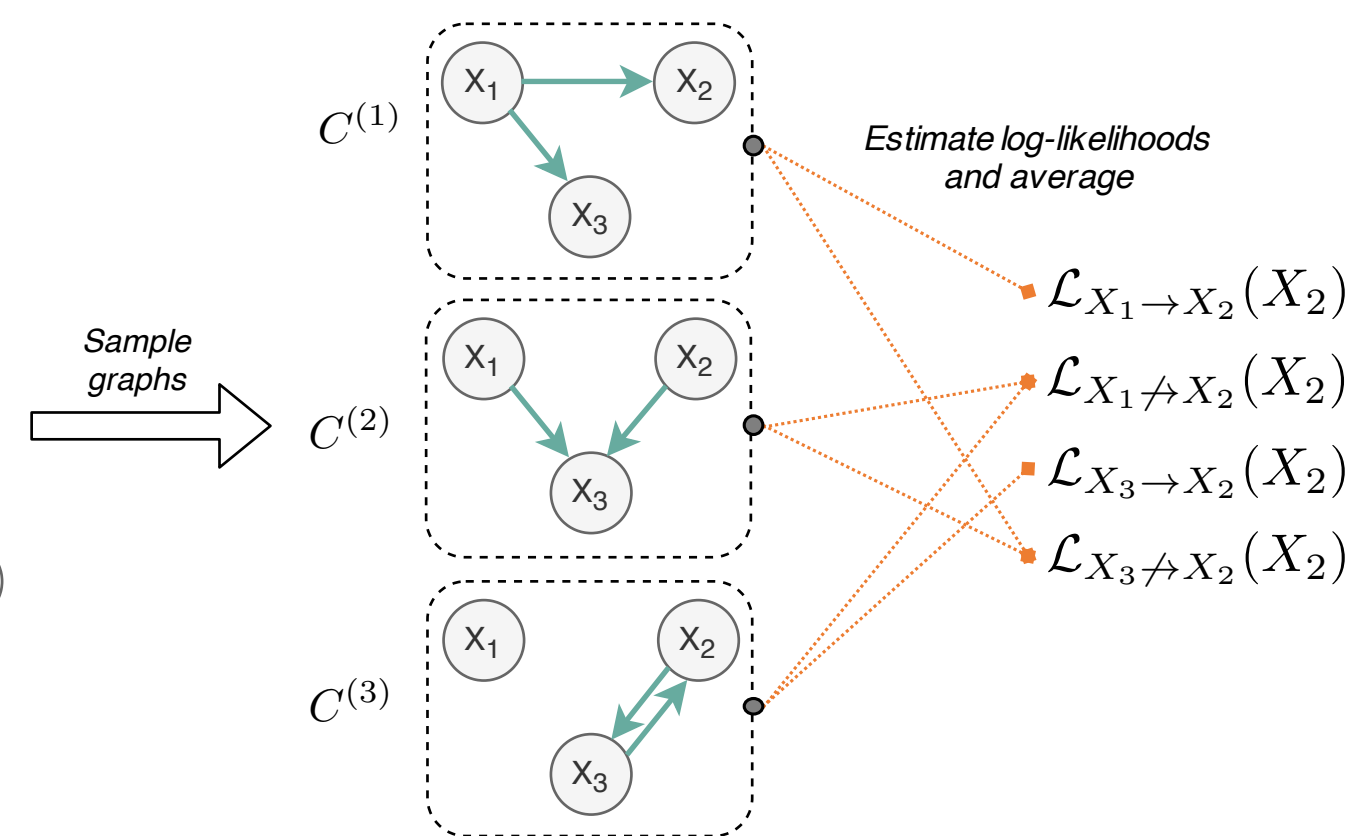
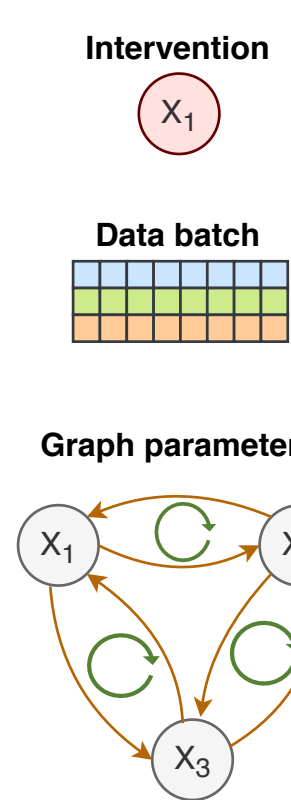


Graph optimization

- Unbiased, low-variance gradient estimator for γ and θ via REINFORCE and Monte-Carlo sampling
- Intuition: sample interventional data and K graphs, and check for each edge whether its existence improved the child's likelihood estimate

$$\frac{\partial}{\partial \gamma_{ij}} \tilde{\mathcal{L}} = \sigma'(\gamma_{ij}) \cdot \sigma(\theta_{ij}) \cdot \mathbb{E}_{\mathbf{X}, C \sim \pi_{ij}} [\mathcal{L}_{X_i \rightarrow X_j}(X_j) - \mathcal{L}_{X_i \not\rightarrow X_j}(X_j) + \lambda_{\text{sparse}}]$$

- Orientations only updated on interventions



$$\begin{aligned} \frac{\partial}{\partial \gamma_{12}} \tilde{\mathcal{L}} &= \dots \\ \frac{\partial}{\partial \gamma_{32}} \tilde{\mathcal{L}} &= \dots \\ \frac{\partial}{\partial \theta_{12}} \tilde{\mathcal{L}} &= \dots \end{aligned}$$

Latent confounders

- Latent confounders between two observable variables without direct causal relation cause a unique pattern in the graph parameters
- An edge between the two variables is disadvantageous on interventional data but beneficial when intervening on any other variable
- Phenomenon can be detected by recording observational and interventional gradients on γ separately, and combine in a score function:

$$\text{lc}(X_i, X_j) = \sigma(\gamma_{ij}^{(O)}) \cdot \sigma(\gamma_{ji}^{(O)}) \cdot (1 - \sigma(\gamma_{ij}^{(I)})) \cdot (1 - \sigma(\gamma_{ji}^{(I)}))$$

Convergence

- Under mild conditions, we can guarantee that ENCO converges to the correct, directed acyclic causal graph despite no acyclic constraints
- Central condition: the expected improvement of the likelihood estimate by adding a correct edge must be greater than the sparsity regularizer λ_{sparse}

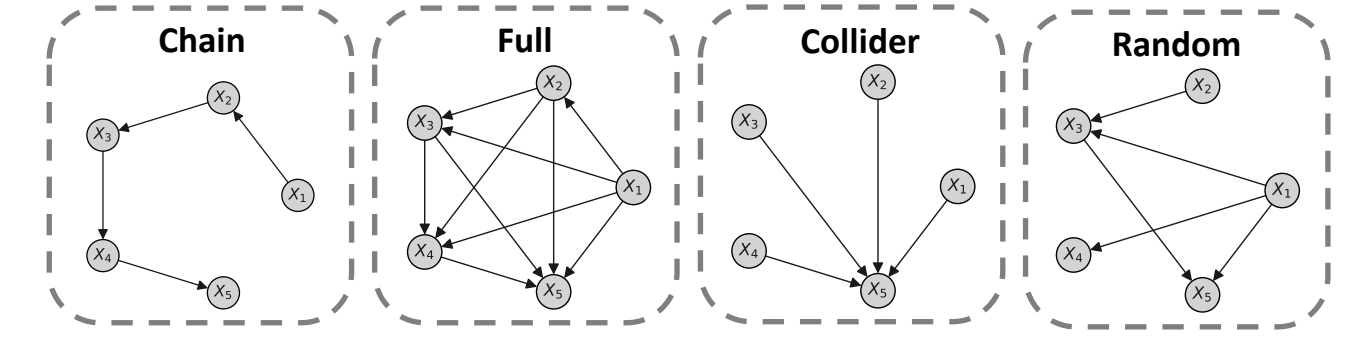
$$\min_{\hat{\mathbf{p}} \subseteq \text{gpa}_i(X_j)} \mathbb{E}_{\hat{I} \sim p_I(I)} \mathbb{E}_{\hat{p}_I(\mathbf{X})} [\log p(X_j | \hat{\mathbf{p}}, X_i) - \log p(X_j | \hat{\mathbf{p}})] > \lambda_{\text{sparse}}$$

- Tradeoff: low values of λ_{sparse} might require longer training times

Experiments

Synthetic graphs

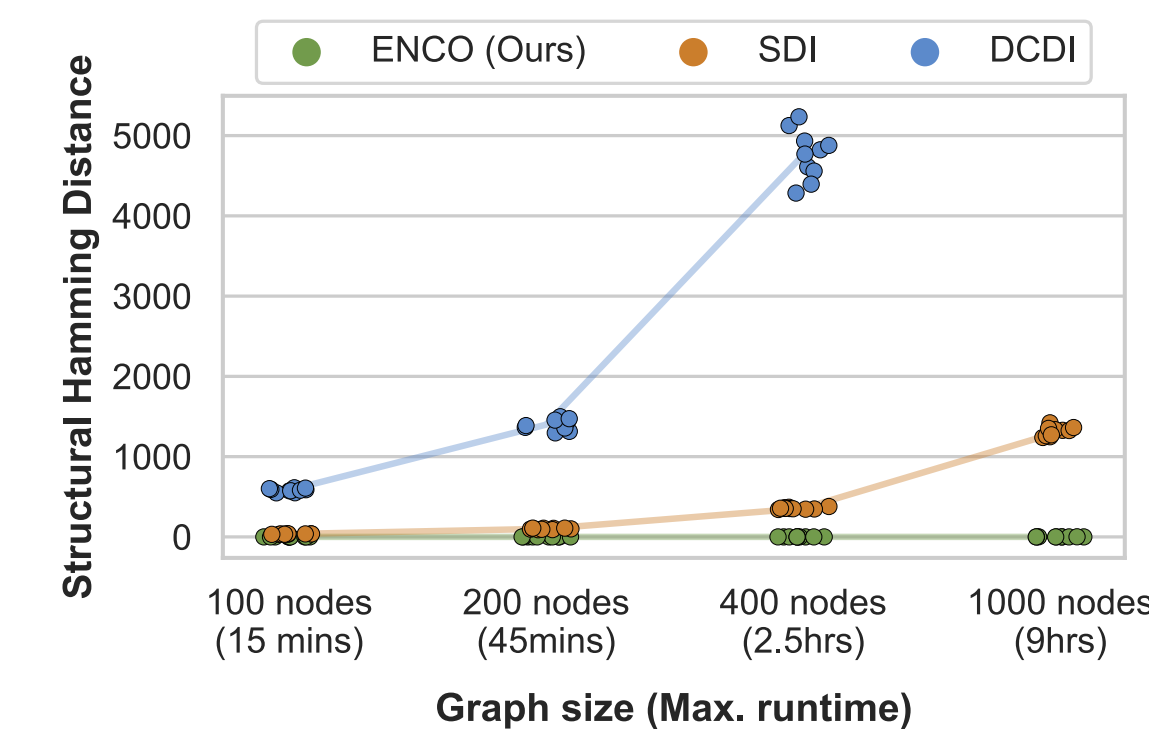
- Synthetically generated graphs with 25 categorical variables of different structures
- ENCO recovers 5 out of 6 without errors



Graph type	bidiag	chain	collider	full	jungle	random
GIES [Hauser and Bühlmann, 2012]	47.4 (± 5.2)	22.3 (± 3.5)	13.3 (± 3.0)	152.7 (± 12.0)	53.9 (± 8.9)	86.1 (± 12.0)
IGSP [Wang et al., 2017]	33.0 (± 4.2)	12.0 (± 1.9)	23.4 (± 2.2)	264.6 (± 7.4)	38.6 (± 5.7)	76.3 (± 7.7)
SDI [Ke et al., 2019]	2.1 (± 1.5)	0.8 (± 0.9)	14.7 (± 4.0)	121.6 (± 18.4)	1.8 (± 1.6)	1.8 (± 1.9)
DCDI [Brouillard et al., 2020]	3.7 (± 1.5)	4.0 (± 1.3)	0.0 (± 0.0)	2.8 (± 2.1)	1.2 (± 1.5)	2.2 (± 1.5)
ENCO (Ours)	0.0 (± 0.0)	0.0 (± 0.0)	0.0 (± 0.0)	0.3 (± 0.9)	0.0 (± 0.0)	0.0 (± 0.0)

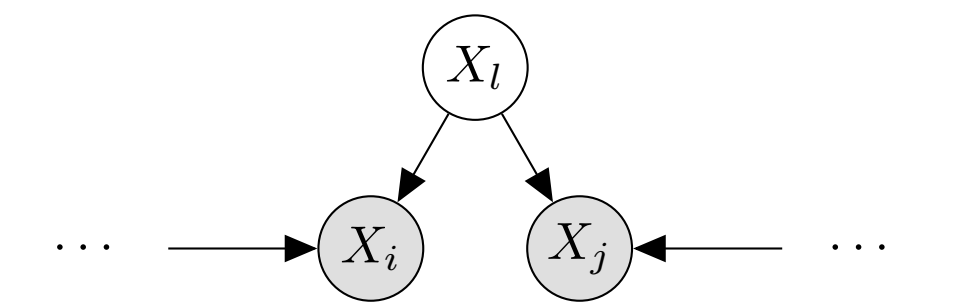
Scalability

- Testing ENCO and strongest baselines on scalability to large, synthetic graphs of 100 to 1000 variables
- With same NN and hardware setting, ENCO is more efficient making only two errors overall in 10 graphs of 1000 variables



Latent confounders

- Synthetic graphs with 25 variables plus 5 latent confounders on arbitrary node pairs except parent-child relations
- ENCO detects most confounders while missed confounders don't effect the SHD



Metrics	ENCO
SHD	0.0 (± 0.0)
Confounder recall	96.8% ($\pm 9.5\%$)
Confounder precision	100.0% ($\pm 0.0\%$)

BnLearn Repository

- Real-world inspired causal graphs from the BnLearn Repository [Scutari, 2010] scaling from 5 to 441 variables with peaked marginal distributions and deterministic variables
- ENCO achieves perfect reconstruction for most graphs, and even works well on *diabetes* with many deterministic variables

Dataset	cancer (5 nodes)	asia (8 nodes)	sachs (11 nodes)	child (20 nodes)	alarm (37 nodes)	diabetes (413 nodes)	pigs (441 nodes)
SDI [Ke et al., 2019]	3.0	4.0	7.0	11.8	24.6	422.4	18.0
ENCO (Ours)	0.0	0.0	0.0	0.0	1.0	2.0	0.0

Takeaways

- Splitting graph parameters into edge existence and orientation for greater control over gradients without acyclicity constraints
- Efficient graph optimization using low-variance gradient estimators by testing generalization to interventions
- Convergence guarantees can be given under mild conditions with respect to likelihood improvements per edge
- ENCO reliably and efficiently recovers causal graphs with up to 1000 variables, including latent confounders

Check out our paper and code for details!